

Interactive Crime Data Visualization Platform

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ABSTRACT

Crime data has been displayed in all sorts of media and it is commonly presented through traditional charts, such as line graphs, bar plots, and scatter plot maps. Moreover, crime information sometimes has the power to provoke diverse emotions, from sadness to curiosity. In this work, we want to appropriate these particularities to present crime data in a novel way. Our goal is to permit the users not only to explore the data but also interact with it in a way they can see details, traditionally not enabled. We gamified the visualization of Chicago's homicide data along with a sonification technique. Data was translated into different sounds, and they were presented to the user in a Unity game application, where they can act as a detective to explore the data. In the era of an overwhelming amount of data, this work is a demonstration of a different way to analyze and present crime information.

KEYWORDS

Insert keyword text, Insert keyword text, Insert keyword text, Insert keyword text

1 Introduction

Crime information was a way to shock and resonate with people at the same time [2][5]. We see it every day how the media channels are full of crime news [13]. They are an interesting selling product since they provoke all kinds of emotions: from sadness to curiosity [1][8]. In many of these news media, Data Visualization plays a key role, bringing scientific and objective information about the displayed crime [3]. Crime Data Visualization supports the given crime information and should be consumed in a clear but almost educative manner.

Although crime data information results in subjective feelings, they are usually presented in an objective and static format, most of them through known charts, graphs, and tables. The compelling output derives from the text following the visualization, the photos or

pictures, or just the crime itself. We are interested in presenting crime data and its nuances in a novel and creative way, where the user can not only be a consumer, but also an explorer.

Munzer describes six ways to analyze data: Discover, Present, Enjoy, Annotate, Record, and Derive [6]. Although this might not be the best description, we are interested in the third approach to displaying crime data. In this project, we want to take advantage of this vast spectrum of feelings resulting whenever we face crime information, or more specifically, crime data.

2 Related Work

Our search demonstrated that not many authors explored feelings or emotions when facing crime data. Or even the consumer of this information interactivity seems to be under-researched. We did find many proposed interactive technologies exploring crime information, but they mainly have the goal to support police work or extract patterns and insights, such as forecasting or network connections.

For this purpose, many works have been developed especially following a spatial analysis. Many projects have been proposed where interactivity was supported by applications such as Google Maps and GIS. Silva, L. et. al. created CrimeVis, an interactive visualization tool to extract insights from crime in the city of Rio de Janeiro, such as time evolution patterns or concentration spatial analysis [7]. ToppiReddy, H et. al. programmed a framework to display crime information in Google Maps tools, where the investigator can visualize hotspots of crime, not only the location, but a 3D street view of the region, filter crime types, and produce data reports [9]. Feng, M. et. al. proposed an application able to handle big data on crime where users can explore maps and get insights through pie charts, time series graphs, word clouds, and scatter plots [4].

In our work, we will focus on crime derived from Chicago's Police public data portal [14]. It already presents an interactive dashboard, where users can explore data from 2001 to the present, filtering different crimes, navigating to Chicago's areas, and building charts such as bar and scatter plots. More on the dataset used will be provided in the following section. Our goal is to present data, where users can interact with it beyond the traditional way, but still be able to get pattern insights.

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3 Dataset

We used Chicago’s dataset and focused specifically on Homicide data. Chicago is considered one of the most dangerous cities in the United States, and the concentration of crimes (especially murders) among neighborhoods is notorious [12]. Our data contained 731 points of homicides with information about the location and details about the crime, from September 2021 to September 2022. Figure 1 shows a sample of a scatter plot of how they are spread in Chicago’s area. In this project specifically, we just used the location information and the column indicating if anyone was arrested or not for that crime.

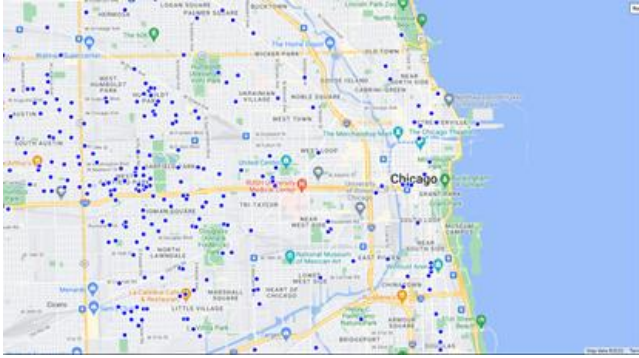


Figure 1: Crime locations (homicide) in a part of Chicago.

4 Motivation and Development of the Creative Process

We started the process of exploring the data to evoke emotions in the user. Our first approach was to use sonification to present the data. Sonification is the process of providing data information through non-speech audio [11]. One of the first registers of this technique dates back to 1908, when the Geiger Counter was invented, a device that detects radiation and outputs a sound accordingly [10]. Although sonification can deliver meaningful information to a database, some visualization scientists do not consider it a good technique, since audio is processed in a sequential stream in our brains, making it hard to understand large sets of data [6]. Our idea was to combine sonification with animated visualization so that not only those with musical understanding could interpret the data but also layman listeners. The decision to use sonification comes from the fact that music provokes all kinds of emotions in the listeners, and we are interested in this relationship between data presentation and cognitive interaction.

This first part of the project consisted of translating crime coordinates into music notes. The idea was that the latitude would give the frequency (pitch) of the note and the longitude would give the time sustained from one note to another. Therefore, we would have acuter sounds at the top part of the map and graver sounds at the bottom. In the same way, if points (or crimes) are close to one another we would listen to short notes, but if they are

apart from one another the note would be sustained longer. We add another feature for the information of arresting from the dataset. If the label was “yes” in the arrest column we would translate it to a piano instrument, and if “no”, we would hear a violin instrument. Now the user not only could have information on the location and concentration of crime but also if there is an arrest involved by just listening to the sound played.

The 731 sounds were played in a sequential way from left to right taking about seven minutes to “read” the map. The program to output the final sound was built in Python, using the SCAMP library, and will be publicly available later.

After feedback and discussion, the next iteration of the project then focused on trying to provide a way for the user to dive deeper into the dataset and introduce a level of interaction and gamification to the data exploration process, while at the same time keeping the data sonification aspect of the original approach intact. The next sections cover this approach, where the idea was to turn the data exploration process into a detective-themed simulation or game to familiarize the user with a crime-based dataset and invoke emotions using sound along with the dataset elements.

5 Methodology of the game design

To introduce a novel way of interacting with the crime dataset, an approach was developed, which would gamify the process of data exploration and interaction. This method would enable the development of a framework, that applied traditional game development practices, to the task of understanding and exploring large datasets, in an effort to introduce a more interactive and involved approach to the process. This method is merely a proof of concept that displays the functionality, and the flow of gamification in the data exploration process.

5.1 Data Wrangling and Cleanup

The original crime database was taken from the Chicago police government website [14]. At first, the latitude and longitude values of the locations were normalized from a scale of 0 to 100. This provided two integer columns, named Grid X (derived from latitude) and Grid Y (derived from longitude), which were used to provide the spatial values of each crime “hotspot” with respect to the generated grid. The formula used for normalized, truncated latitude and longitude values was:

$$= \text{TRUNC}((\{value\} - \{minimumvalue\}) / (\{maximumvalue\} - \{minimumvalue\}) * 100, 0)$$

Then, another formula was used to detect duplicate values of {Grid X, Grid Y} together, and based on that, the first value was kept, while the others were removed. This was done to ensure that each tile has only one unique crime “hotspot”. Even though the system supports multiple crime “hotspots” visually, it is a more complicated process to keep track of such data in real time and is beyond the scope of this proof-of-concept. The duplicate detection formula was:

Insert Your Title Here

WOODSTOCK'18, June, 2018, El Paso, Texas USA

$$= AND(EXACT(\{Grid X, Grid Y\}_{TOWN}, \{Grid X, Grid Y\}_{TOWN+1}))$$

These operations resulted in a clean dataset, void of NULL values, and each tile location having one “hotspot” as well as preserving at least one record from the same location, from the original dataset. The final number of unique crime entries was 588, which ended up becoming 588 crime “hotspots” for the simulation.

5.2 Tools used

The simulation has been developed using the game engine Unity 2019.3.4f1. The primary programming language used was C#. Microsoft Excel was used to clean up the data and convert the dataset into a text file. The text file contains Comma Separated Values that follows a specific syntax and order, which is exactly how it is set up in the excel file.

5.3 Simulation design

The simulation was divided into four states: LOAD, PLAY, INSPECT, and REVIEW. These four states reside in the Game Manager, which can be the central core of the simulation, and have specific roles to play before and during the simulation runtime. This section further expands upon these states and details the inner workings of the simulation.

5.3.1 LOAD state

This state is responsible for loading the text file that contains the comma-separated values, related to the crime dataset. This state handles all the pre-processing necessary for the simulation to parse the values, and then create individual classes for each crime data. These classes are termed “Hotspots” and each “Hotspot” contains information related to a crime event. For example, this version contains details about the Case Number, Approximate Location, Arrest Made, Type of Crime, etc. All these fields exist as columns in the initial dataset; thus, this class can be considered a “digital container” for crime records. In addition to that, this class also stores an additional sound clip variable, that contributes towards the gamification of the experience. Once each crime record has been copied into a container, these containers are stored inside a list item. Thus, before the simulation even begins, the data is already loaded in and stored for usage, saving up on processing power and resources during runtime. This is also the state where the user can set the size of the grid (i.e., the number of tiles in the X and Y direction).

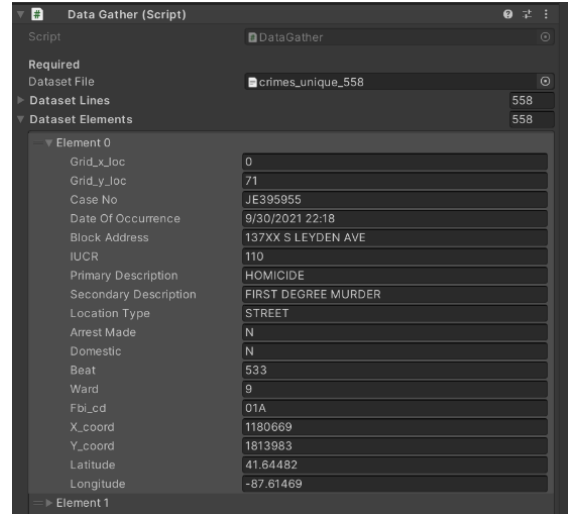
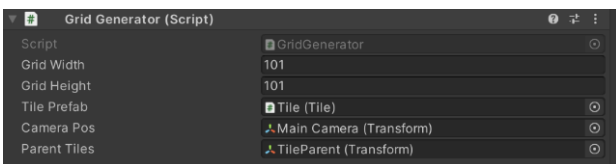


Figure 2: Generating a 101x101 tile grid and using 558 data values from the cleaned dataset.

5.3.2 PLAY state

This is the main stage of the simulation, where the user will engage with the data, and start driving toward the objective of the simulation. The first step during this state is for the Grid Manager system to load in a specified size grid, set in the LOAD state. The grid tiles are set to then hold the containers for each crime hotspot data. These are also responsible for detecting the mouse pointer’s location (i.e. which tile the pointer collides with). Once this first state is done, the player avatar (icon) is spawned in, and the user can now control its placement. The player has a limited field of view. This means that at any given moment, taking the player as the center, the user can move their mouse pointer to any edge, and they can “look ahead” a certain number of tiles. This ensures that there is not an overload of data, and the camera does not move too far away from the player.

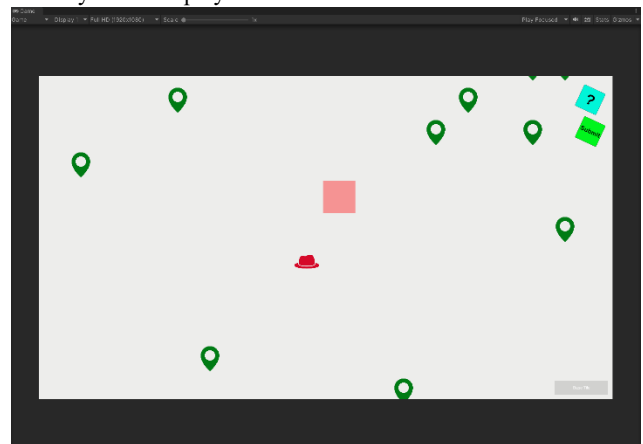


Figure 3: Main play screen (PLAY state); The red hat is the player icon, and the green marks are the crime “hotspots” containing the information from the dataset.

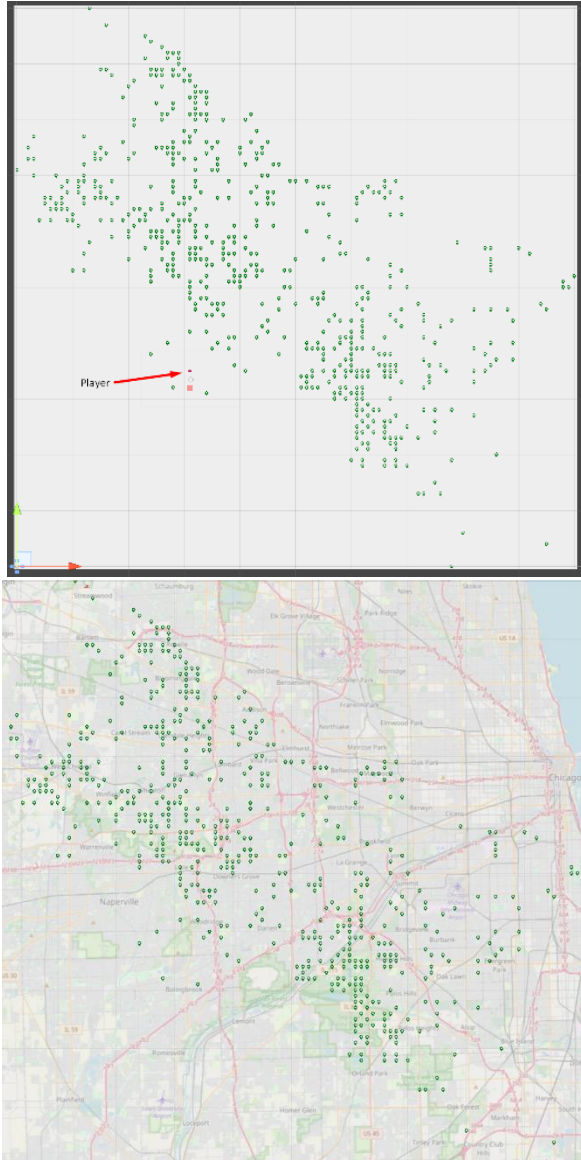


Figure 4: Location of the crime “hotspots” in the simulation and overlaid on top of the Chicago and neighboring areas

Once the user clicks on the player icon, two scenarios are possible:

- If the clicked-on tile is empty, the player icon “moves” to that location, indicating that there is no crime information in that tile.
- If the clicked-on tile has crime data, the simulation enters a new state (Section 6.3.3), and then, they can dive deeper into the data that exists for that specific tile (representing a crime location).

During this stage, there is another feature available to the user. There exists a “Show Hint” button on the screen, that is developed specifically for giving clues to find a specific crime hotspot, during the current session of the simulation. Once this button is clicked,

the INSPECT state is active (6.3.3) and the player is free to read the hint. This has been done to enable the aspect of a gameplay loop, where the user has a set objective to accomplish during the session. Gamification aspects like these will enable the user to stay engaged with the data, treat the cumbersome task of looking at datasets, and potentially turn it into an enjoyable experience.

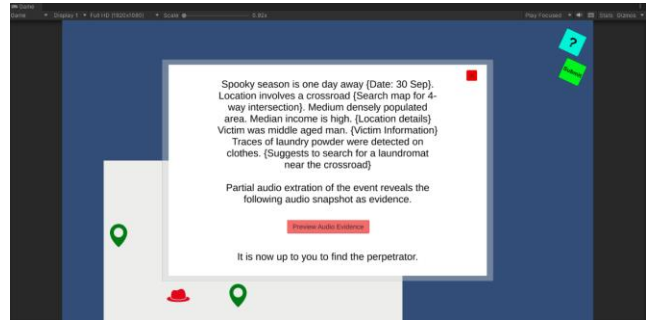


Figure 5: An example of a hint given to the player (the information in the curly brackets is supposed to be what the hint alludes to).

The main idea behind this stage is to present the data to the user, so they can navigate it freely. The primary reason for this is now the data exists in its own digital interactable space, instead of inside an application like Excel. This gamified aspect of data exploration caters to the agency of the user and puts them in control of how they want to go through this journey of understanding the data present in front of them.

5.3.3 INSPECT state

This is the state that fires up, when the user (after clicking on their icon), clicks on a tile that contains a crime “Hotspot”. This hotspot is basically a container for all the information related to the crime data in the dataset. The grid has been laid out in a way that it overlays itself on a specific section of the Chicago map (for which the dataset is associated with), thus each tile having the crime location from that approximate real-world location. In the inspect state, a new panel pops up that lays out all the information from the dataset, in the form of a card with fields for each of the columns in the dataset. Some additional buttons and panels will contribute to the gameplay loop of the simulation (such as a button that plays the specific sound at each location, a general image that relates in some way to the hint, etc.) The INSPECT state is important since this is the analysis and decision-making part of the gameplay loop, where the user is expected to take all the information provided in the hint, and conclude, whether they have found the correct crime “hotspot” or not. During this state, the user cannot interact with the grid map or other “hotspots” without closing the panel, thus going back to the PLAY state.

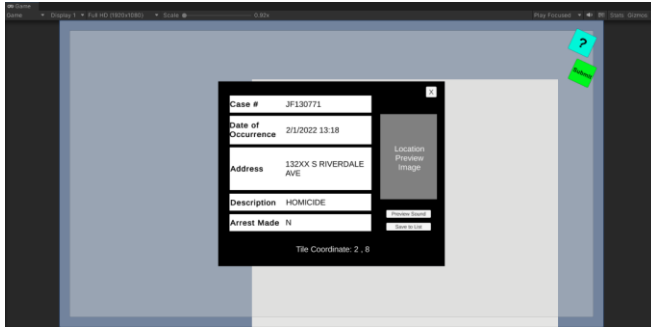


Figure 6: The INSPECT state with a “hotspot” card with the details about the crime.

5.3.3 REVIEW state

The final state of the simulation is the REVIEW state. It is like INSPECT state, however, this is executed when the user is sure that they have found the correct “hotspot” containing the details hinted towards by the given clue (and has the same sound clip as the clue). This marks endgame, where the player either successfully finds the specific “hotspot” and can restart, or finds the wrong one, and continues on their journey. This state is where the user is provided feedback for their choices.

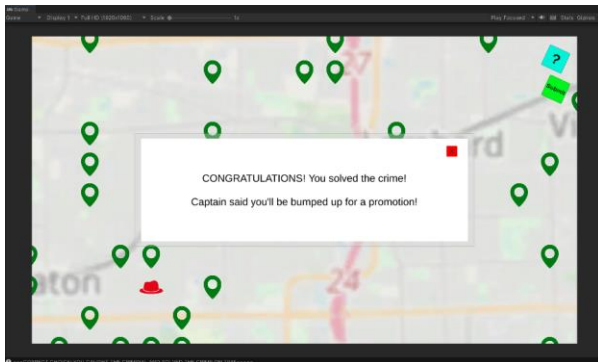


Figure 7: If the user successfully finds the location with the same sound clip as given in the clue.

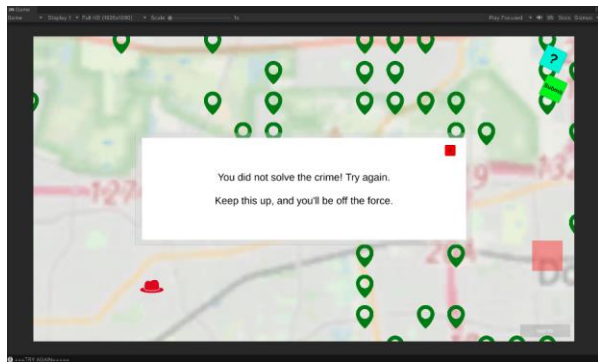


Figure 8: If the user is unsuccessful in finding the correct location with the same sound clip as given in the clue.

6 Conclusion

To conclude, this was an attempt to reimagine the way that crime data is interacted with by users and explorers when going through the process of visualizing data. By introducing concepts of gamification, an approach could be taken where the understanding of the data is intrinsic, and the user starts seeing patterns and diving deeper, just by interacting with how the data is presented to them. Another aspect of it is the sonification, where now instead of just stimulating the visual sensory mode, the audial sensory mode is also stimulated and thus each element of data within the exploration process leaves a bigger mark on the user. The goal was to not only leverage insights from the data, but also potentialize emotions typical of this sort of data.

7 Limitations and Future Work

Since this is a proof of concept, the main idea behind this simulation was to test whether this concept of gamifying and sonification of data was even possible. This framework works for the basic purpose but there do exist a few limitations. First of all, there does not exist a standard format for recording crime data, thus the code is highly tailored for the dataset. A more generalized approach needs to be developed for this simulation to have a greater effect size. Next, due to problems beyond the scope of this project, each tile was limited to having only one crime “hotspot”. Reality is very different, and an ideal approach would be to extend the existing system to display multiple “hotspot” cards, contained in the same location. Third, there exists only a single layer of depth to the crime data. It would be better if there were more layers and data manipulation avenues available to the user, to build out a narrative for themselves (like an actual detective), and derive insight from it. Having data from external sources would also be a major addition.

Finally, another avenue we would like to explore is to develop a better sonification aspect, and create a standardized channel and marks system for it, so that there is a sense behind the sound clip associated with each “hotspot”. This would enhance the gameplay design of the simulation, and enable the user to make a better judgment based on a lot of details and evidence, that they can freely manipulate and find dependencies and links.

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